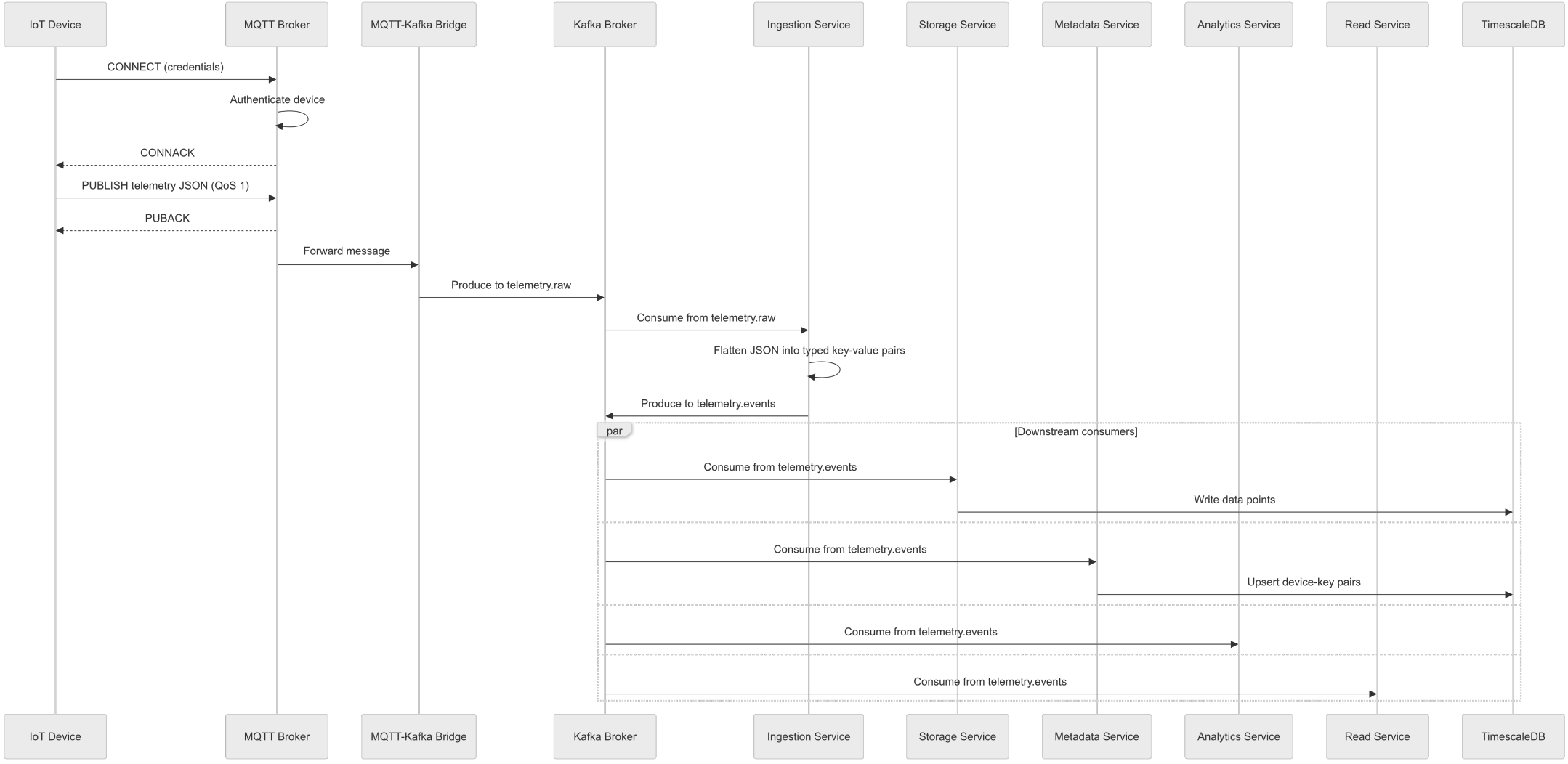
IoT Device
MQTT Broker
MQTT-Kafka Bridge
Kafka Broker
Ingestion Service
Storage Service
Metadata Service
Analytics Service
Read Service
TimescaleDB
CONNECT (credentials)
Authenticate device
CONNACK
PUBLISH telemetry JSON (QoS 1)
PUBACK
Forward message
Produce to telemetry.raw
Consume from telemetry.raw
Flatten JSON into typed key-value pairs
Produce to telemetry.events
par
[Downstream consumers]
Consume from telemetry.events
Write data points
Consume from telemetry.events
Upsert device-key pairs
Consume from telemetry.events
Consume from telemetry.events